

Real-Time Distributed Keyword Spotting

Scheduling Analysis and Precision Effects in a
Three-Node NLP Pipeline

Armaan Amatyia | Syed Hasan Shamsi
University of Houston, TX, USA

INTRODUCTION

- > Voice-controlled systems are critical in robotics, automation, and safety-critical environments.
- > Emergency stop commands require near-instantaneous recognition and action.
- > AI models, specifically deep learning, introduce unpredictable execution times.
- > **Motivation:** Can these AI-based systems meet strict real-time guarantees?



WHAT IS NATURAL LANGUAGE PROCESSING(NLP)?

The Core Concept

NLP allows computers to interpret human language across text and speech. Used in Siri, Alexa, and industrial command interfaces.

Key Idea

Human language is converted into numerical representations. Models use learned patterns to predict meaning and trigger actions.



THE NLP INFERENCE PIPELINE



Audio Capture

Captures raw audio signal and converts it into numerical features that the model then understands.



Wav2Vec2 Model

Designed for speech processing. It takes the captured features then predicts what keyword was spoken



Action Trigger

Prediction triggers immediate command execution under sub-150ms real-time constraints.

SYSTEM ARCHITECTURE

Node A

Audio acquisition & feature
extraction from 16kHz
stream.

Node B

Core AI inference running
Wav2Vec2 on an NVIDIA
RTX 5060 Ti GPU.

Node C

Responsible for validating
command, safety logic,
and watchdog timer
actuator control.



SCHEDULING MODEL (RMS)

$$U \geq 1$$

UTILIZATION FACTOR

- To analyze the system, we model it as a set of periodic tasks and use Rate Monotonic Scheduling.
- RMS assigns priorities based on how frequently tasks run.
- Using theoretical calculations, we computed system utilization, and in our case, it exceeds 1.
- This means that according to Real time scheduling theory, the system should not be schedulable.

$$U = \sum_{i=1}^n \frac{C_i}{T_i} \leq n (2^{1/n} - 1)$$

Prediction: The system should fail to meet deadlines.

EXPERIMENTAL SETUP

Component	Specification
Software Stack	Python, PyTorch, HuggingFace
AI Processor	NVIDIA RTX 5060 Ti GPU
Precision Modes	FP32 (Standard) vs FP16 (Reduced)
Network	UDP (Lightweight Protocol)



RESULTS (FP32 vs FP16)



Surprising Finding: FP16 optimization actually increased average latency by 9% and nearly doubled worst-case latency.

CASCADE EFFECTS

Node B Delay

When the AI model experiences latency spikes, results are transmitted late, violating periodic expectations.

Node C Impact

Delays propagate, causing the safety validation node to miss its own reception deadlines.

Overrun Stats

FP16: 12 downstream misses observed.

FP32: 0 misses (stable performance).

| CONCLUSION & KEY TAKEAWAYS



- > **FP16:** Not a reliable optimization for single-sample Transformer inference in real-time.
- > **Performance:** The system met the 150ms deadline with a substantial margin.
- > **Evaluation:** Real-world performance differs significantly from theoretical predictions.
- > **Final Word:** Empirical testing is essential for AI in safety-critical systems.

Future Works

- > TensorRT INT8
Quantization.
- > VxSim RTOS
Deployment.
- > UDP Network
Impairment Testing
via UDP proxy.

QUESTIONS?

Thank you!

armaanamatya2014@gmail.com | hasanshamsi91@gmail.com